

Technology Stack

SMARTBRIDGE - Technology Stack

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-6970
- **Team Size:** 3
- **Team Leader:** Prince Kumar P
- **Team Members:** Gowtham L, Guru Moorthi M
- **Project Name:** AI FitTrack
- **Maximum Marks:** 2 Marks

Technology Stack Details

S.No	Architecture Component / Layer PDF	Technology Chosen PDF	Justification / Purpose PDF
1	Frontend / Client-Side	Postman / Thunder Client / React.js	Provides a flexible client interface to execute HTTP REST API calls and render workout JSON payloads.
2	Backend / Server-Side	Node.js & Express.js	Lightweight, non-blocking asynchronous event loop allowing fast execution of API endpoints and middleware.
3	Database / Data Storage	MongoDB Atlas & Mongoose ODM	Flexible NoSQL document database ideal for dynamic workout schemas and high performance data queries.
4	Cloud / Hosting / Deployment	MongoDB Atlas Cloud & Local Node Execution	Provides auto-scaling cloud database hosting with IP whitelisting and reliable connection URIs.

5	Version Control & CI/CD	Git & GitHub	Enables seamless team collaboration, code history tracking, and repository management.
6	Third-Party APIs / Other Tools	Google Gemini API (@google/genai)	Enables AI workout plan generation (gemini-2.5-flash) and personalized fitness insight analytics.